WWII ANTISUBMARINE AIRCRAFT

Aircraft played a decisive role in anti-submarine warfare during the Battle of the Atlantic from 1940–43. The protection of merchant ships bound for Britain from attack by German U-boats was crucial to the country's survival. Although antisubmarine duties were also undertaken by shipborne aircraft, the major burden was carried by long-range aircraft flying from coastal bases. Under the circumstances, Britain gave surprisingly low priority to RAF Coastal Command, which was responsible for Atlantic antisubmarine patrols. Advocates of strategic bombing wanted long-range aircraft allocated to the bombing offensive against Germany, resenting the diversion of resources to help keep shipping lanes open. One of the major requirements in antisubmarine warfare aircraft was range. Until late 1942 the Allies had no aircraft patrolling the mid-Atlantic, and this gap in convoy air defense was ably exploited by German U-boats. The gap was closed by the introduction of a maritime version of the B-24 Liberator, the Consolidated PB4Y-1, which had a range of around 2,800 miles (4,500km). New equipment made aircraft effective U-boat killers by the end of 1942. They contributed greatly to the effective defeat of the U-boat menace in 1943.

LONG-RANGE CATALINA
One of the most familiar aircraft of its time, the durable, dependable "Cat" had a distinguished service record in many theaters of war with both the American and Allied air forces.

Consolidated PBY Catalina

The PBY Catalina was the US Navy's antisubmarine patrol aircraft from 1936. RAF Coastal Command also ordered large numbers to complement its own Sunderland. More Catalinas were built (over 4,000) than any other flying boat in history. Though slower and less well-armed than the Sunderland, it was tough and adaptable.

Engine 2 x 1,200 hp P&W R-1830 Twin Wasp air-cooled radials
Wingspan 104ft (31.7m) **Length** 63ft 10in (19.5m)
Top speed 196mph (314kph) **Crew** 7
Armament 5 x .5in machine guns; 4,000lb (1,614kg) of torpedoes, depth charges, or bombs

Consolidated PB4Y-1 Liberator

In November 1941, the RAF was the first to use the Liberator for very long-range antisubmarine patrols. A US Navy PB4Y-1 Liberator sunk its first U-boat in November 1942. The Navy had almost 1,000 PB4Y aircraft including a special remote-controlled version designed to attack the V 1 missile sites in France. In August 1944, Lt. Joseph Kennedy, brother of the future US President, was killed in one such operation.

Engine 4 x 1,200hp P&W R-1830-65 Twin Wasp 14-cylinder radials
Wingspan 110ft (33.3m) **Length** 67ft 3in (20.5m)
Top speed 279mph (449kph) **Crew** 9–10
Armament 8 x .5in machine guns; 12,800lb (3,628kg) bombload

Fairey Swordfish Mk.III

The Swordfish, nicknamed the "Stringbag," was so successful as the Royal Navy strike aircraft that, despite its antiquated appearance, it outclassed its intended replacement and served throughout the War. In November 1940, 21 Swordfish torpedo carriers and bombers from the carrier HMS *Illustrious*, sank the Italian fleet at Taranto, for the loss of only one aircraft. From 1943, the Swordfish Mk. III was equipped with radar and operated from convoy escort carriers in the North Sea and Atlantic, sinking many U-boats.

Engine 750hp Bristol Pegasus 30 air-cooled 9-cylinder radial
Wingspan 45ft 6in (13.9m) **Length** 35ft 8in (10.9m)
Top speed 138mph (224kph) **Crew** 2–3
Armament 2 x .303in Vickers machine guns; 1 x torpedo or 1,500lb (681kg) mines, bombs or depth charges, 8 x 60lb (27kg) rockets

Lockheed (A-28/A-29) Hudson I

Engine 2 x 1,200hp Wright R-1820 Cyclone 9-cylinder radials
Wingspan 65ft 6in (20m) **Length** 44ft 4in (13.5m)
Top speed 246mph (396kph) **Crew** 5
Armament 5 x machine guns; 750lb (340kg) bombload

Developed in 1938 as a maritime reconnaissance bomber from the Model 14 Super Electra airliner, the portly Hudson was the first American-built aircraft in RAF service during WWII. Other firsts included the first RAF aircraft to destroy an enemy aircraft and the first to sink a U-boat with rockets. Nearly 3,000 Hudsons served both Allied air forces to the end of the war.

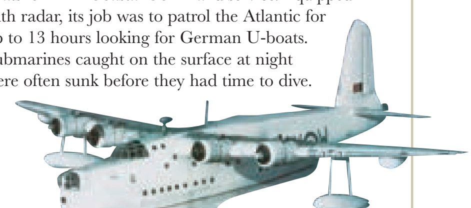
Martin PBM Mariner

Designed in 1937, four years after the more famous Catalina, the Model 162 Mariner flying boat, with its gull wing and distinctive canted tail fins, finally entered service in 1941. Over 1,300 were built in a number of variants, the principal wartime version being the PBM.

Engine 2 x 1,700hp Wright R-2600-12 Cyclone 14-cylinder radials
Wingspan 118ft (36m) **Length** 80ft (24.4m)
Top speed 198mph (319kph) **Crew** 7–8
Armament 7 x machine guns; 4,000lb (1,814kg) bombload

Short S.25 Sunderland

The Sunderland was developed by the Short brothers from their famous commercial "Empire" class flying boats for RAF Coastal Command service. Equipped with radar, its job was to patrol the Atlantic for up to 13 hours looking for German U-boats. Submarines caught on the surface at night were often sunk before they had time to dive.



Engine 4 x 1,200hp P&W Twin Wasp air-cooled 14-cylinder radials
Wingspan 112ft 10in (34.4m) **Length** 85ft 4in (26m)
Top speed 213mph (343kph) **Crew** 10
Armament 12 x .303in Browning machine guns, 2 x .5in machine guns in beam positions; 4,960lb (2,250kg) bombload or depth charges